

Mobile Computing

Subject Code: CS40011 (L-T-P-Cr: 3-0-0-3)

Course Objectives:

- To understand the fundamentals of mobile communication
- To understand the architecture of various Wireless Communication Networks
- To understand the significance of different layers in mobile system Course

Contents:

UNIT I: Introduction to Wireless Networks, Applications, History, Simplified Reference Model, Wireless transmission, Frequencies, Signals, Antennas, Signal propagation, Multiplexing, Modulation, Spread spectrum, Cellular Systems: Frequency Management and Channel Assignment, Types of handoff and their characteristics.

UNIT II: MAC, Motivation, SDMA, FDMA, TDMA, CDMA, Telecommunication Systems, GSM: Architecture- Location tracking and call setup, Mobility management, Handover, Security, GSM, SMS, International roaming for GSM, Call recording functions, Subscriber and service data management, DECT, TETRA, UMTS, IMT-2000.

UNIT III: Wireless LAN, Infrared vs. Radio transmission, Infrastructure, Adhoc Network, IEEE 802.11 WLAN Standards, Architecture, Services, HIPERLAN, Bluetooth Architecture & protocols.

UNIT IV: Mobile Network Layer, Mobile IP, Dynamic Host Configuration Protocol, Mobile Transport Layer, Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/Fast recovery, Transmission/Time-out freezing, Selective retransmission, Transaction Oriented TCP.

UNIT V: WAP Model, Mobile Location based services, WAP Gateway, WAP protocols, WAP user agent profile caching model, Wireless bearers for WAP, WML, WML Scripts – WTA – iMode, SyncML.

Course Outcomes: Upon completion of the course, the students will be able to

CO1: Perceive the fundamentals of mobile communication

CO2: Describe the architecture of various Wireless Communication Network

CO3: Recognize different layers in mobile system

CO4: Develop a strong grounding in the fundamentals of mobile Networks

CO5: Apply knowledge in MAC, Network, and Transport Layer protocols of Wireless Network

CO6: Comprehend and develop a lightweight network stack

Textbooks: 1. Jochen Schiller, “Mobile Communication”, Second Edition, Pearson Education, 2008.

2. Theodore, S. Rappaport, “Wireless Communications, Principles, Practice”, Second Edition, Pearson, 2023.

Reference Books: 1. C. Siva Ram Murthy, B. S. Manoj, “Adhoc Wireless Networks: Architectures and Protocols”, Second Edition, Pearson Education, 2008.

2. Vijay. K. Garg, “Wireless Communication and Networking”, Morgan Kaufmann Publishers, 2007.